

Zhao Du

PH.D. APPLICANT IN BIOMEDICAL ENGINEERING, ROBOTICS, AND HUMAN-MACHINE INTERFACES

Shenzhen, Guangdong, China

✉ zhao.du@siat.ac.cn | Google Scholar

Education

University of Chinese Academy of Sciences; Shenzhen Institutes of Advanced Technology, Chinese Academy of Sciences

Shenzhen, China

M.ENG. IN CONTROL ENGINEERING; GPA: 3.6/4.0

Sep. 2024 - Expected Jun. 2027

- Graduate training at SIAT, a Chinese Academy of Sciences research institute with interdisciplinary programs in engineering, biomedical sensing, and intelligent systems; UCAS ranked No. 54 globally in the 2025-2026 U.S. News & World Report Best Global Universities ranking.

Beijing Jiaotong University, School of Electronic and Information Engineering

Beijing, China

B.ENG. IN AUTOMATION; GPA: 3.2/4.0

Sep. 2019 - Jun. 2023

- Completed engineering coursework in control, signal processing, embedded systems, sensors, algorithms, machine learning, and optimization.
- Beijing Jiaotong University is a national Double First-Class university with engineering strengths in transportation, automation, and information systems.

Publications

Democratizing Precision: High-Performance Hand Gesture Recognition via Low-Cost Exoskeleton and Physics-Informed Graph Neural Networks

IEEE Transactions on
Instrumentation and Measurement

FIRST AUTHOR; PUBLISHED

- Introduced a low-cost hand-motion sensing pipeline with a 3-D-printed exoskeleton and a physics-informed graph network that encodes hand-joint structure and biomechanical constraints.
- Demonstrated strong cross-user generalization and data efficiency, supporting affordable gesture-recognition systems for wearable interfaces, rehabilitation training, and assistive devices.

Cross-Subject Hand Gesture Recognition via Inter-Modal Alignment of sEMG and A-mode Ultrasound for Subject-Invariant Robotic Control

IEEE/RSJ International Conference
on Intelligent Robots and Systems
(IROS)

FIRST AUTHOR; ACCEPTED

- Proposed an inter-modal alignment framework that learns subject-invariant representations from sEMG and A-mode ultrasound, addressing user-to-user signal shift in multimodal physiological sensing.
- Validated the method in offline evaluation and robotic-control experiments, showing practical value for calibration-light assistive human-machine interaction.

Online Hand Gesture Recognition with Hybrid sEMG-AUS Sensing and a Spiking Neural Network for Robotic Applications

IEEE Robotics and Automation
Letters

FIRST AUTHOR; INVITED FOR SECOND-ROUND REVIEW

- Developed a dual-branch spiking neural network that jointly processes electrophysiological and morphological features from hybrid sEMG-AUS sensing for online hand gesture recognition.
- Targets low-latency and energy-efficient embedded deployment, providing a practical path toward wearable prosthetic interfaces and robotic-hand control.

Experience

CRRC Zhuzhou Locomotive Co Ltd, Urban Rail Development Department

Zhuzhou, China

ENGINEER; URBAN RAIL DEVELOPMENT AND ENGINEERING DOCUMENTATION

Aug. 2023 - Jun. 2024

- Tasked with supporting technical documentation, interface coordination, and engineering drawing review for urban rail vehicle development.
- Prepared routine engineering records and review materials, coordinating feedback among mechanical, electrical, and systems engineering teams.
- Served as a documentation and workflow-oriented engineer, using CAD/UG tools and spreadsheet automation to organize design files, compare versions, and improve recurring review workflows.

NIO, Assisted and Intelligent Driving Division

Beijing, China

DATA ENGINEERING INTERN; DATA PIPELINE AND TOOLING CONTRIBUTOR

Mar. 2025 - Jul. 2025

- Tasked with supporting large-scale engineering data analysis, quality checking, and visualization for intelligent-driving development workflows.
- Built Python/Linux tools for data retrieval, statistics, configuration updates, and result visualization to improve routine data-inspection efficiency.
- Worked as a tooling-oriented engineering contributor, connecting data requirements with practical scripts and reproducible analysis workflows.

Leadership

School of Artificial Intelligence, University of Chinese Academy of Sciences

Beijing, China

PRESIDENT OF THE GRADUATE STUDENT UNION

2024 - 2025

- Led a 30-member graduate student union serving approximately 500 students, coordinating the publicity, sports and arts, and academic departments.
- Organized the “Artificial Intelligence+” academic salon series, coordinating speakers, promotion, venues, and student participation for interdisciplinary discussion among graduate students.
- Coordinated a cross-school evening gala with four schools, approximately 1,000 audience members, and about 150 student performers, managing program review, rehearsal scheduling, and on-site execution.
- Planned school-level sports and community activities, including badminton selection and inter-team matches, to support student engagement and cross-cohort communication.

Awards

2025	National Scholarship , Ministry of Education of China	China
2025	Outstanding Student , University of Chinese Academy of Sciences	China
2025	Outstanding Student Leader , University of Chinese Academy of Sciences; selected among the top 5% of student leaders	China
2022	Second Prize , Lanqiao Cup Algorithm Competition, Beijing Division	China

Skills

Programming and Robotics	Python, ROS, C/C++
Learning Models	Graph networks, spiking neural networks